

7. The table shows some properties of elements in Period 3 of the Periodic Table.

| Element | Melting point (°C) | Boiling point (°C) | Density (g/cm <sup>3</sup> ) | Appearance   | Malleability | Conductivity  |
|---------|--------------------|--------------------|------------------------------|--------------|--------------|---------------|
| Na      | 98                 | 882                | 1.00                         | shiny solid  | malleable    | good          |
| Mg      | 650                | 1091               | 1.75                         | shiny solid  | malleable    | good          |
| Al      | 660                | 2470               | 2.70                         | shiny solid  | malleable    | good          |
| Si      | 1410               | 3265               | 2.35                         | shiny solid  | brittle      | semiconductor |
| P       | 44                 | 281                | 1.80                         | red solid    | brittle      | poor          |
| S       | 113                | 444                | 2.05                         | yellow solid | brittle      | poor          |
| Cl      | −101               | −34                | 0.003                        | green gas    | n/a          | poor          |

(a) One of the elements is difficult to classify as a metal or non-metal.

Identify this element and give your reasoning.

[2]

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- (b) Tick (✓) **two** boxes which correctly describe the change in density and boiling point for the elements across Period 3. [2]

The density of metals and non-metals increases

☐

The boiling point of metals increases but the boiling point of non-metals shows no trend

☐

The density of metals shows no trend but the density of non-metals decreases

☐

The boiling point of metals and non-metals shows no trend

☐

The density of metals increases but the density of non-metals shows no trend

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The boiling point of metals shows no trend but the boiling point of non-metals decreases

☐

The density of metals decreases but the density of non-metals shows no trend

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- (c) Argon is the next element in Period 3 after chlorine, Cl.

State why it is not possible to predict a melting point for argon using the information in the table. [1]

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(d) Phosphorus is found in phosphoric acid,  $\text{H}_3\text{PO}_4$ .

(i) During the production of phosphoric acid, phosphorus is heated to  $60^\circ\text{C}$ .

Give the state of phosphorus at  $60^\circ\text{C}$ . Explain your choice.

[2]

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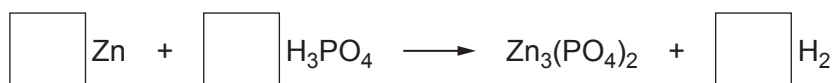
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(ii) Phosphoric acid reacts with zinc to produce zinc phosphate and hydrogen.

Balance the equation for this reaction.

[1]



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